

Solution

ELECTRICITY -10

Class 10 - Science

Section A

1. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
2. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Since in the given case the voltage is same, therefore, $H = \frac{V^2}{R}t = \text{constant}$. Hence, if R is halved t must be halved.
3. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
4. **(c)** A is true but R is false.
Explanation: Silver is a good conductor of electricity but it is not used to make electric wires because it is expensive.
5. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Resultant resistance is the addition of individual present in the series circuit. So, according to the above statement,
Resultant resistance $(R) = R_1 + R_2 + R_3$. So total resistance will be $3R$. In parallel combination, resultant resistance will be $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$.
Here, both assertion and reason are true and reason is the correct explanation of assertion.
6. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Electric shock is due to the electric current flowing through a living body. When the bird perches on a single high power line, no current passes through its body because its body is at equipotential surface, i.e, there is no potential difference. While when man touches the same line, standing barefoot on ground, the electrical circuit is completed through the ground. The hands of man are at high potential and his feet are at low potential. Hence large amount of current flows through the body of the man and therefore, gets a fatal shock.
7. **(c)** A is true but R is false.
Explanation: Resistivity is a material property. And Resistance is inversely proportional to the cross-section area of the wire. So, the thinner the wire, the higher the resistance.
8. **(d)** A is false but R is true.
Explanation: Since the bulbs are joined in series, so when one bulb is removed from chain then the resistance of the chain is decreased, hence current flowing through each bulb is increased. As heat produced $\propto i^2$, hence light gets increased in the room.
9. **(c)** A is true but R is false.
Explanation: It creates an open circuit in the series, which interrupts the flow of electrical current to all the bulbs. In general, higher wattage bulbs emit more light because they use more electricity to generate that light.
10. **(c)** A is true but R is false.
Explanation: A is true but R is false.
11. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: When two resistance R_1 and R_2 connected in parallel then their equivalent resistance will be $r = \frac{R_1 R_2}{R_1 + R_2}$.

12. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
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15. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Most of the part of consumed power of bulb is dissipated as heat but very less part gets converted into light. So, there is a need for thermal isolation in order to reduce heat losses. This is the reason why bulbs are filled with inactive nitrogen and argon. So, both assertion and reason is true and reason is the correct explanation of assertion.
16. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
17. (c) A is true but R is false.
Explanation: The conductor has a positive charge, so it has lost two electrons.
18. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
19. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Metals are good conductors of electricity. It is because of the presence of a large number of free electrons in metals. And for metals electrons are the main cause for thermal conduction. That's why all good conductors of heat are also good conductors of electricity.
20. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
21. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
22. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: $\text{Glow} = \text{Power}$
 $(P) = I^2 R$
 $\frac{dP}{P} = 2 \left(\frac{dI}{I} \right) = 2 \times 0.5 = 1\%$
23. (c) A is true but R is false.
Explanation: Ohm's law states that the electric current flowing through a metallic wire is directly proportional to the potential difference across its two ends. The expression is written as:
 $V = IR$
 Here R (resistance of the wire) is constant value then only the statement will be valid.
 $V \propto I$ only if $\frac{V}{I} = \text{constant}$
24. (c) A is true but R is false.
Explanation: $\text{Conductivity} = \frac{1}{\text{Resistivity}}$
 $\text{Conductivity} \times \text{resistivity} = 1$
25. (c) A is true but R is false.
Explanation: Fuse wire must have high resistance because according to Joules law $H = I^2 R t$, heat produced is high if R is high. The melting point must be low so that wire may melt with increase in temperature. As a current larger than the specified value flows through the circuit the temperature of the fuse wire increases. This melts the fuse wire and breaks the circuit.
26. (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Volume of material on stretching remains same, i.e, $Al = \text{constant}$. When l becomes three times, its area of cross-section (A) becomes $\frac{1}{3}$ rd. From $R = \rho \frac{l}{A}$, R would become 9 times.

27. (a) Both A and R are true and R is the correct explanation of A.

Explanation: According to the formula: $R = \rho \frac{l}{A}$

Where, R : resistance

ρ : resistivity

l : length of conductor

A : area of cross-section of conductor

Here resistivity of the material never varies, if length is also kept constant. When area is halved then resistance of the material gets halved as the resistance depends on 3 factors: length, area and nature of the material.

28. (a) Both A and R are true and R is the correct explanation of A.

Explanation: Resistance of the connecting wires is much smaller than the electric appliances to which current is supplied by the wires.

29.

- (d) A is false but R is true.

Explanation: In tube light majority portion of radiation comes under the visible region while bulb radiation consists of visible, ultraviolet, infrared radiation giving less visible part.

30.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Power loss $= i^2 R = \left(\frac{P}{V}\right)^2 R$ [P = Transmitted power]

31.

- (c) A is true but R is false.

Explanation: ρ is the characteristic of the material of resistors. It does not depend on the length and cross-sectional area of resistors. But R depends on the length and the cross-sectional area of the resistor. So, R_1 may be greater than R_2 even when.

32.

- (c) A is true but R is false.

Explanation: A is true but R is false.

33.

- (c) A is true but R is false.

Explanation: Here, $P = \frac{E^2}{R}$, so $P \propto R$ only when I is constant.

Here I increase as R is decreased. Hence the reason is wrong.

34.

- (b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

35.

- (c) A is true but R is false.

Explanation: Here for using the alloys in making toasters, electric iron is based on only one reason for resistivity. If the resistivity of the material is high which is being used to make these items, then heat dissipation will be more. And these items are used for heating only. So, the reason is wrong.

So, assertion is true, but the reason is false.

36.

- (c) A is true but R is false.

Explanation: As $P = VI$, hence for the transmission of same power, high voltage implies less current. Therefore heat energy losses ($H = I^2 R t$) are minimized if power is transmitted at high voltage.

37.

- (c) A is true but R is false.

Explanation: In the case of a short-circuited battery, the current

$$I = \frac{E(\text{e.m.f. of the battery})}{r(\text{internal resistance})} \neq 0$$

Terminal voltage, $V = IR = I(i) = I(0) = 0$

Where, $R = \text{external resistance} = 0$

38.

(c) A is true but R is false.

Explanation: A is true but R is false.

39.

(c) A is true but R is false.

Explanation: A is true but R is false.

$$\rho = \rho_0(1 + \alpha \Delta T)$$

40.

(a) Both A and R are true and R is the correct explanation of A.

Explanation: Both A and R are true and R is the correct explanation of A.

41.

(c) A is true but R is false.

Explanation: A is true but R is false.

42.

(a) Both A and R are true and R is the correct explanation of A.

Explanation: In a conductor, there are large numbers of free electrons. When we close the circuit, the electric field is established instantly with the speed of electromagnetic waves which causes electron drift at every portion of the circuit. Due to which the current is set up in the entire circuit instantly. The current which is set up does not wait for the electrons to flow from one end of the conductor to another end. It is due to this, the bulb glows immediately when the switch is on.

43.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation:

- i. At high temperatures, metal wires have a greater chance of short-circuiting because both resistance and resistivity of a metal wire varies with temperature.
- ii. Resistance is the property of a substance by virtue of which it opposes the flow of current through it is known as the resistance.
- iii. It is directly proportional to temperature.
- iv. While resistivity is an intrinsic property do not depend on size and shape but do depend on temperature for metals it increases with an increase in temperature.

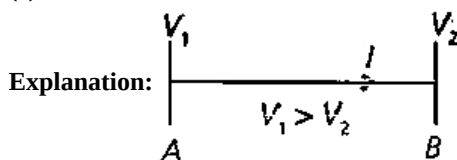
44.

(a) Both A and R are true and R is the correct explanation of A.

Explanation: Voltage measured in the parallel circuits are always equal. As all the parallel circuits start from one point and end at another point and always the potential difference between the two points will always be the same. So, this is the reason why voltmeter is always connected in parallel across the circuit. Thus, both assertion and reason are true and reason is the correct explanation of assertion.

45.

(c) A is true but R is false.



Suppose A and B are two regions having potentials V_1 and V_2 such that $V_1 > V_2$. So the electric current will flow from A to B (i.e., from higher potential to lower potential). Since electrons move opposite to the direction of current, hence, the electrons move from a region of lower potential to a region of higher potential.

46.

(c) A is true but R is false.

Explanation: A is true but R is false.

47.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: Both A and R are true but R is not the correct explanation of A.

48. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Heater wire must have high resistance and high melting point, because in series current remains same, therefore according to Joule's law, $H = I^2Rt$, heat produced is high if R is high melting point must be high, so that wire may not melt with increase in temperature.
49. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
50. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Current will not flow when the two bodies are at the same potential. When their charges are same, their potential may be different. Hence current may flow in this case.
51. **(d)** A is false but R is true.
Explanation: A is false but R is true.
52. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: When a conductor is placed in an electric field, it is said that electric current flows through the conductor due to the flow of electrons from positive to negative terminals. This is the reason for flowing electric current. If the circuit gets opened, then the path of flow of electrons gets interrupted and current stops flowing. Thus, this option is the correct answer.
53. **(c)** A is true but R is false.
Explanation: Assertion is true but the reason is false.
 As heat flow is dependent on three factors:
 i. Current flowing through it.
 ii. Time period of flow of current.
 iii. Resistance of the conductor.
 So, the reason is false that heat flow only depends on square of current. As it is dependent on several other factors too.
54. **(d)** A is false but R is true.
Explanation: It is quite clear that in a battery circuit, the point of lowest potential is the negative terminal of the battery and the current flows from higher potential to lower potential.
55. **(c)** A is true but R is false.
Explanation: It is a common error to say that $V = Ri$ is a statement of Ohm's law. The essence of Ohm's law is that the value of R is independent of the value of V. The equation $V = Ri$ is used for finding the resistance of all conducting devices, whether they obey Ohm's law or not.
56. **(d)** A is false but R is true.
Explanation: The current in a wire is due to flow of free electrons in a definite direction. But the number of protons in the wire at any instant is equal to number of electrons and charge on electrons is equal and opposite to that of proton. Hence, net charge on the wire is zero.
57. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Resistance of wire $R = \rho \left(\frac{l}{A} \right)$
 Where ρ is the resistivity of material which does not depend on the geometry of wire. Since when the wire is bent, resistivity, length, and area of cross-section do not change, therefore the resistance of wire also remains the same.
58. **(c)** A is true but R is false.
Explanation: A is true but R is false.

Section B

59. (a) (i) - (c), (ii) - (b), (iii) - (d), (iv) - (a)

Explanation:

- A Battery maintains a potential difference between two points in an electric circuit. It is made up of two or more cells connected together.
- In a series combination of resistors, the same current flows through each resistor.
- The current is the same in every part of the circuit. In a parallel combination of resistors, the current divides at a junction.
- A Fuse protects electrical appliances from high voltage. It is a safety device consisting of a strip of wire that melts and breaks an electric circuit if the current exceeds a safe level.

60. (a) (i) - (a), (ii) - (c), (iii) - (b), (iv) - (d)

Explanation:

- The SI unit of electric current is ampere (symbol A). It is named after the French scientist - Andre Marie Ampere.
- The ohm (symbol Ω) is the SI unit of electrical resistance, named after German physicist - Georg Simon Ohm.
- The SI unit of potential difference is volt (symbol V), named after the Italian physicist - Alessandro Volta.
- The SI unit of resistivity is ohm-metre (symbol $\Omega\text{-m}$).

61.

(d) (i) - (d), (ii) - (a), (iii) - (c), (iv) - (b)

Explanation:

- Tungsten is exclusively used for making filaments of electric bulbs.
- Nichrome is an alloy of nickel, chromium, manganese, and iron and does not oxidize easily at high temperatures. It is commonly used in electrical heating devices like electrical iron, heaters, etc.
- Superconductors are materials that conduct electricity with zero resistivity.
- EMF (Electro Motive Force) of a battery creates a potential difference across a circuit.

62. (a) (i) - (b), (ii) - (d), (iii) - (a), (iv) - (c)

Explanation:

- **Electric current** (I) is the amount of charge (Q) flowing through an area of the cross-section in-unit second.
- **Electric Potential** at a point is the work done in moving a unit charge from infinity to that point.
- **Resistance** (R) is a property that resists the flow of current in a circuit. According to Ohm's law, the current (I) flowing through a conductor is directly proportional to the potential difference (V) applied across the ends of the conductor.
- **Power** (P) is the rate at which electric energy is lost (dissipated) or consumed in an electrical device due to resistance (R).

63.

(c) (i) - (b), (ii) - (d), (iii) - (a), (iv) - (c)

Explanation:

- Silver is the best conductor. It has a resistivity of 1.6×10^{-8} ohm-m.
- Ebonite is a very hard rubber obtained by vulcanizing natural rubber for prolonged periods. It has a resistivity in the range of 10^{15} to 10^{17} ohm-m and acts as an insulator.
- Silicon is a semi-conductor.
- Constantan is an alloy of copper and nickel and is used to make standard resistors. It has a resistivity of 49×10^{-6} ohm-m.

64.

(c) (i) - (c), (ii) - (b), (iii) - (d), (iv) - (a)

Explanation:

- $I = \frac{Q}{t}$
- $P = VI = I^2R = \frac{V^2}{R}$
- $F = \frac{Kq_1q_2}{r^2}$ where K is a constant of proportionality.
- $R = \rho \frac{l}{A}$ where ρ is a constant of proportionality and is called the electrical resistivity of the material of the conductor.

65.

(c) (i) - (a), (ii) - (c), (iii) - (b), (iv) - (d)

Explanation:

- An Ammeter (from Ampere Meter) is used to measure electric current (flow of electric charge) in an electric circuit.
- A Galvanometer is an electromechanical instrument for detecting and measuring small electric currents.
- A Battery maintains a potential difference between two points in an electric circuit.
- A Rheostat is an electrical instrument used to control current in a circuit by varying the resistance.

66.

(d) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Explanation: (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

67. (a) (i) - (d), (ii) - (a), (iii) - (c), (iv) - (b)

Explanation:

- When several resistors are joined in series, the equivalent resistance of the combination is equal to the sum of the individual resistances.

$$R_{\text{series}} = R_1 + R_2 + R_3 + \dots$$
- When several resistors are joined in **parallel**, the reciprocal of the equivalent resistance of the combination is equal to the **sum of the reciprocals of the individual resistances**.

$$\frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$
- When several resistors are joined in **parallel**, the potential difference across each resistor and across the combination is the same.

$$\frac{V^2}{R_{\text{parallel}}} = \frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3} + \dots \text{ OR } P_{\text{parallel}} = P_1 + P_2 + P_3 + \dots$$
- When several resistors are joined in **series**, the same current flows through each resistor. $I^2 R_{\text{series}} = I^2 R_1 + I^2 R_2 + I^2 R_3 + \dots$ OR $P_{\text{series}} = P_1 + P_2 + P_3 + \dots$
 Please note that **power is additive** for any configuration of a resistive circuit.

68.

(c) (i) - (d), (ii) - (a), (iii) - (c), (iv) - (b)

Explanation: $W = VQ$

$\therefore 1 \text{ Volt} \times 1 \text{ Coloumb} = 1 \text{ Joule} \dots (i)$

$P = VI$

$\therefore 1 \text{ Volt} \times 1 \text{ Ampere} = 1 \text{ Watt} = 1 \text{ Joule per sec} \dots (ii)$

From (i) and (ii),

$1 \text{ Joule} = 1 \text{ Watt} \times 1 \text{ sec} = 1 \text{ Volt} \times 1 \text{ Coloumb}$ or $W \times s = V \times C$

$\frac{\text{Coulomb}}{\text{Second}}$ corresponds to $\frac{Q}{t}$.

Section C

69.

(d) Resistance

Explanation: Power is inversely proportional to resistance if the potential difference is constant. Power can directly be proportional to the resistance.

70.

(b) All of these

Explanation: All of these

71.

(d) More

Explanation: More

72.

(a) Do a lot of work on each charge it encounters.

Explanation: The electric potential difference or voltage of a battery is the potential energy difference across its terminals for every Coulomb of charge. A high voltage battery maximizes this ratio of energy/charge by doing a lot of work on each charge it encounters.

Section D

73. (a) Variable resistance or rheostat

Explanation: The symbol  represents a variable resistance or a rheostat. It is used to change the resistance in an electric circuit.

74.

(d) (ii)

Explanation: The circuit shown in figure (ii) is properly connected.

In other figures, the connections are not correct.

For example: in figure (i), the voltmeter is connected in series and not in parallel. Similarly in figure (iii), the ammeter is in parallel which should be in series. In figure (iv), the negative terminal of the battery has been connected to the positive terminal of ammeter whereas it should be connected to the negative terminal and not positive terminal of the ammeter.